Species-Aware Module Review: mcl-PHA monomer supply from de novo fatty-acid synthesis in *Pseudomonas putida* KT2440 (UniPathway UPA00212)

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module bucket:** unipathway:UPA00212 — PhaG bridge from de novo fatty-acid synthesis (FAS) to medium-chain-length polyhydroxyalkanoate (mcl-PHA) monomers **Candidate gene(s):** phaG — PP_1408 | O85207 | annotated as (R)-3-hydroxydecanoyl-ACP:CoA transacylase (EC 2.4.1.-)

Database cross-check (UniProt REST, Iteration 2). O85207 is a **reviewed Swiss-Prot** entry: gene `phaG`, locus `PP_1408`, organism *P. putida* KT2440 (taxon 160488), name "(R)-3-hydroxydecanoyl-ACP:CoA transacylase", EC 2.4.1.-, mapped to **UniPathway UPA00212**, KEGG `ppu:PP_1408`, BioCyc `G1G01-1499-MONOMER`. Its FUNCTION field records only the 1998 in-vitro transacylase assay and does **not** incorporate the 2012 thioesterase revision — the legacy annotation is entrenched at the Swiss-Prot level and inherited downstream. The candidate 3HA-CoA ligase `PP_0763` = **Q88PT5**, **TrEMBL (unreviewed)**, submission name "Medium-chain-fatty-acid CoA ligase", **no EC number** and no curated 3-hydroxy specificity.

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1. Executive summary

- The single candidate gene, **phaG (PP_1408 / O85207)**, is a **genuine, experimentally characterized enzyme of the target strain** *P. putida* KT2440. It is the historically defined "metabolic link" between de novo fatty-acid synthesis and mcl-PHA biosynthesis (Rehm et al. 1998, P 9727022). This module step is **covered** by direct target-organism evidence.

- **Curation-critical caveat:** the metadata annotation "3-hydroxyacyl-ACP:**CoA transacylase** (EC 2.4.1.-, acyltransferase GO)" reflects the **original 1998 model, which has since been biochemically revised**. Current evidence (Wang et al., *Appl. Environ. Microbiol.* 2012, 78:519–527; corroborated by P 25732207, P 30799088) indicates **PhaG functions as a (R)-3-hydroxyacyl-ACP thioesterase** (EC 3.1.2.-), releasing **free (R)-3-hydroxy fatty acids** rather than directly forming an acyl-CoA.
- Consequently, **complete monomer supply to a CoA thioester (the substrate of PhaC) requires a second enzyme: a (R)-3-hydroxyacyl-CoA ligase, encoded in KT2440 by PP_0763** (EC 6.2.1.-). This gene is **absent from the module metadata** but is mechanistically required and present in the target genome.
- **Recommendation:** mark the PhaG step **covered (with a revised-activity note)**; flag the module as **module_needs_revision** because a required 3HA-CoA ligase step (PP_0763) is missing; treat the EC 2.4.1.- / acyltransferase GO mapping as a **likely over-propagated legacy annotation**.

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2. Target-organism pathway definition

Included process. The channelling of intermediates of **de novo fatty-acid biosynthesis** (ACP-bound (R)-3-hydroxyacyl groups, principally C8–C12) toward **mcl-PHA monomer supply**. In *P. putida* KT2440 this is the route used when cells grow on **structurally unrelated carbon sources** (gluconate, glucose, sugars, styrene-derived acetyl-CoA), typically under **nitrogen/nutrient limitation**, when no exogenous fatty acids are available.

Mechanistically (revised model): 1. FAS produces **(R)-3-hydroxyacyl-ACP**. 2. **PhaG** removes the 3-hydroxyacyl group from ACP — now understood as a **thioesterase**, yielding **free (R)-3-hydroxy fatty acid**. 3. A **(R)-3-hydroxyacyl-CoA ligase (PP_0763)** activates the free acid to **(R)-3-hydroxyacyl-CoA**. 4. (Downstream, **outside this module**) **PhaC1/PhaC2 synthases** polymerize the CoA monomers into mcl-PHA.

Neighboring processes to keep separate. - **PhaC polymerization** (PP_5003 phaC1 / PP_5005 phaC2) — the terminal step; correctly excluded from UPA00212. - **β-oxidation-fed monomer supply via PhaJ** (R-specific enoyl-CoA hydratase) — the alternative route used when growing

on fatty acids; a distinct monomer-supply module, correctly excluded. - **Fatty-acid de novo synthesis proper** (fab genes) and **rhamnolipid synthesis (RhlA/RhlG)**, which compete for the same 3-hydroxyacyl-ACP pool — overview/neighboring maps, not part of this bucket.

Alternate names / database definitions. phaG gene product is variously annotated as "(R)-3-hydroxydecanoyl-ACP:CoA transacylase," "3-hydroxyacyl-ACP-CoA transferase," or "3-hydroxyacyl-ACP thioesterase" (PhaG). EC has historically been left as **2.4.1.-** (glycosyl/acyl transferase placeholder); the revised activity better fits a **thioesterase (EC 3.1.2.-)**.

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3. Expected step model

#	Expected step	Enzyme / activity	Gene in KT2440	Status
1	FAS supplies (R)-3-hydroxyacyl-ACP	fatty-acid synthase (fab)	fabB/fabF/ fabG etc.	Upstream context (not in module)
2	Release of 3-hydroxyacyl group from ACP	**PhaG** — 3-hydroxyacyl-ACP thioesterase (revised) / historically ACP:CoA transacylase	**phaG / PP_1408**	**Covered (activity revised)**
3	Activation of free (R)-3-hydroxy fatty acid to CoA thioester	(R)-3-hydroxyacyl-CoA ligase (EC 6.2.1.-)	**PP_0763**	**Gap in metadata / likely present**
4	Polymerization of 3HA-CoA to mcl-PHA	PHA synthase	phaC1/ phaC2	Out of scope (downstream)

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4. Candidate genes and evidence

phaG — PP_1408 | O85207 — HIGH CONFIDENCE (identity), REVISED ACTIVITY

- **Likely role:** Removes the (R)-3-hydroxyacyl moiety from FAS-derived acyl-ACP to feed mcl-PHA monomer supply.

- **Evidence type — direct, target strain (strong):** Cloned from *P. putida* KT2440 by phenotypic complementation of PHA-deficient mutants; 295-aa / 33,876 Da protein; 44% identity to RhlA; transcriptionally induced on gluconate; heterologous expression conferred PHA synthesis from gluconate in a non-producer (*P. oleovorans*). Purified enzyme showed 3-hydroxyacyl-CoA↔ACP acyl-transfer activity in vitro (Rehm et al. 1998, P 9727022).
- **Pathway-role confirmation (strong):** Five PHA-negative KT2440 Tn5 mutants were all complemented by phaG → PhaG is the **only** FAS→PHA link in *P. putida* (Rehm, Mitsky, Steinbüchel 2001, P 11425728).
- **Curation-relevant caveats:**
- **Activity revised** from transacylase to **thioesterase** (Wang et al. 2012, via P 25732207). The metadata EC 2.4.1.- and "acyltransferase activity" GO are **legacy/over-propagated**; the enzyme does not by itself yield acyl-CoA.
- **PhaG alone is insufficient** for CoA-monomer supply/PHA in heterologous hosts unless a 3HA-CoA ligase is co-supplied — repeatedly observed in *E. coli* engineering (P 25732207, P 30799088). This is the practical signature of the thioesterase mechanism.
- Homolog (>99% identity) in *P. putida* CA-3 shows the same nitrogen-limitation-dependent expression (P 16085828); PhaG(Pa) in *P. aeruginosa* is a functional ortholog (P 10713430) — good genus-level transfer, but KT2440 has its own direct evidence, so transfer is not needed here.

PP_0763 — (R)-3-hydroxyacyl-CoA ligase (UniProt Q88PT5) — NOT IN METADATA, SHOULD BE ADDED

- **Likely role:** ATP-dependent ligase activating free (R)-3-hydroxy fatty acid → (R)-3-hydroxyacyl-CoA, completing monomer supply for PhaC.
- **Evidence type — direct, target strain (moderate–strong):** PP_0763 gene product from *P. putida* KT2440 shown to have (R)-3HA-CoA ligase activity (Wang et al. 2012, cited in P 25732207); a *P. aeruginosa* paralog (PA3924) provides the same activity and outperforms PP_0763 in a recombinant system (P 25732207).
- **Database status (curation-relevant):** UniProt **Q88PT5**, **TrEMBL/unreviewed**, named "Medium-chain-fatty-acid CoA ligase", **no EC number assigned**, no explicit 3-hydroxyacyl specificity in the record — under-curated relative to the experimental evidence.
- **Curation note:** Should be **promoted to full fetch-gene review** and added as a required step if the module is intended to represent supply of **CoA** monomers (PhaC substrate). If the module is deliberately scoped only to "release from ACP," PP_0763 belongs to an adjacent activation step.

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5. Gaps, ambiguities, and likely over-annotations

- **Over-annotation (likely):** phaG EC **2.4.1.-** ("transacylase"/acyltransferase). This is the 1998 interpretation; the 2012 biochemistry favors **thioesterase (EC 3.1.2.-)**. GO "acyltransferase activity" is correspondingly questionable; a hydrolase/thioesterase GO term may be more accurate. Flag for GO-curation review.
- **Metadata gap:** the **3HA-CoA ligase step (PP_0763)** required to reach the PhaC substrate is missing. Without it the module cannot, on its own, deliver a CoA-thioester monomer — a satisfiability gap for the "monomer supply" intent.
- **Ambiguity / paralogy:** whether a single dedicated 3HA-CoA ligase or several acyl-CoA synthetases (broad EC 6.2.1.-) fulfil step 3 in KT2440 is not fully resolved; acyl-CoA ligase families are large and prone to broad EC mapping and cross-annotation.

- **Boundary correctness:** the module correctly excludes PhaC and the β-oxidation/PhaJ route. The FAS-route boundary itself is well supported (PhaG is the sole FAS→PHA link).

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6. Module and GO-curation recommendations

Item	Recommendation
PhaG / PP_1408 step	**covered** — but attach a note that the true activity is **3-hydroxyacyl-ACP thioesterase**, not ACP:CoA transacylase.
EC / GO mapping for phaG	**module_needs_revision / GO request** — replace/augment EC 2.4.1.- with EC 3.1.2.- (thioesterase); re-evaluate "acyltransferase activity" GO vs. a thioesterase/hydrolase term.
3HA-CoA ligase step (PP_0763)	**gap** in current metadata — add as a required activation step (EC 6.2.1.-) for CoA-monomer supply, or explicitly document that the module stops at ACP release.
Module boundary vs. PhaC and β-oxidation/PhaJ	**keep separate** — current scoping is correct.
Overall module satisfiability	**candidate_uncertain / module_needs_revision** — the sole listed step is present, but the module as written mis-states the enzyme's activity and omits a mechanistically required ligase step.

No step should be marked `not_expected_in_target_taxon`: both PhaG and a 3HA-CoA ligase are present and functional in KT2440.

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7. Genes to promote to full review

1. **phaG / PP_1408 (O85207)** — promote to full `fetch-gene` review to correct the activity/EC/GO annotation (transacylase → thioesterase).
2. **PP_0763** — promote to full `fetch-gene` review; candidate (R)-3-hydroxyacyl-CoA ligase that should likely be added to the module. Check for additional acyl-CoA synthetase paralogs that could share this role.

8. Key references

- Rehm BHA, Krüger N, Steinbüchel A. (1998) A new metabolic link between fatty acid de novo synthesis and PHA synthesis. The phaG gene from *Pseudomonas putida* KT2440 encodes a 3-hydroxyacyl-ACP–CoA transferase. *J Biol Chem* 273:24044–24051.

  P 9727022

  — original characterization in the target strain.
- Rehm BHA, Mitsky TA, Steinbüchel A. (2001) Establishment of the transacylase (PhaG)-mediated pathway for PHA biosynthesis in *E. coli*; PhaG is the only FAS→PHA link in *P. putida*. *Appl Environ Microbiol* 67:3102–3109.

  P 11425728

  — module boundary support.
- Hoffmann N, Steinbüchel A, Rehm BHA. (2000) The *P. aeruginosa* phaG gene product. *FEMS Microbiol Lett* 184:253–259.

  P 10713430

  — genus ortholog.
- Hokamura A, et al. (2015) Biosynthesis of P(3HB-co-3HA) by recombinant *E. coli* from glucose. *J Biosci Bioeng* — states PhaG "actually functions as a 3-hydroxyacyl-ACP thioesterase" and PP0763 has 3HA-CoA ligase activity (citing Wang et al. 2012).

  P 25732207

  — activity revision + PP_0763.
- Goto S, et al. (2019) LA-based polymers containing mcl-3HA in recombinant *E. coli*. Uses "(R)-3-hydroxyacyl-ACP thioesterase (PhaG)" plus "(R)-3-hydroxyacyl-CoA ligase."

  P 30799088

  — corroborates two-enzyme monomer supply.
- O'Leary ND, et al. (2005) PHA from styrene in *P. putida* CA-3; nitrogen-limitation-dependent phaG expression, >99% identity to KT2440.

  P 16085828

- Wang Q, et al. (2012) *Appl Environ Microbiol* 78:519–527 — primary source for PhaG thioesterase activity and PP_0763 3HA-CoA ligase (cited secondarily; recommend direct retrieval during full gene review).

Evidence status & confidence (curation quick-reference)

Claim	Confidence	Evidence type
PP_1408 = phaG = O85207 = the FAS→PHA enzyme in KT2440	**High**	Direct, target strain (P 9727022) + UniProt Swiss-Prot cross-check
PhaG is the sole FAS→PHA link in *P. putida*	**High**	Direct genetics, target species (P 11425728)
PhaG's true activity is thioesterase, not transacylase	**Moderate–High**	Primary source Wang et al. 2012 (AEM 78:519–527), cited secondarily by two independent groups (P 25732207 P 30799088); no dissenting reports found
PP_0763 (Q88PT5) supplies 3HA-CoA ligase activity	**Moderate**	Target strain, reported via secondary citation; UniProt entry unreviewed, no EC
EC 2.4.1.- / acyltransferase GO is over-propagated legacy	**High**	Confirmed entrenched in Swiss-Prot O85207; contradicted by thioesterase mechanism

Note: The pivotal Wang et al. 2012 paper (*Appl. Environ. Microbiol.* 78:519–527) is cited here through two independent secondary sources; direct retrieval of the primary article is recommended during the full `fetch-gene` review of phaG and PP_0763 to confirm the exact assayed activities and EC assignments.

Uncertainty & species-transfer statement

The core annotations (PhaG identity, role as sole FAS→PHA link, revised thioesterase activity, PP_0763 ligase activity) are **direct for *P. putida* KT2440**. Ortholog evidence from *P. aeruginosa* / *P.* sp. 61-3 is supportive but not required for the target-strain conclusions. The main open experimental question is the precise in vivo enzyme(s) and EC/GO term for the ACP-release and CoA-activation steps in KT2440, best resolved by direct review of Wang et al. 2012 and biochemical/structural data for PP_1408 and PP_0763.
